

SIMATS ENGINEERING
ASSESSMENT TOOL 2- COMPARATIVE ANALYSIS

Course Code:	CSA12	Course Title:	Computer Architecture
CO Assessed:	CO4	Assessment:	ASSESSMENT TOOL2- COMPARATIVE ANALYSIS
Weightage:	30%	Total Marks:	25
CO4 Statement:	<i>Implement hierarchical memory systems by differentiating cache levels (L1, L2, L3), virtual memory with paging, and advanced memory technologies (DRAM, SRAM, NAND Flash, MRAM, RRAM) based on latency, bandwidth, and throughput characteristics.</i>		
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ANNEXURE A
COMPARATIVE ANALYSIS

1. Compare L1, L2, and L3 cache levels in terms of latency, bandwidth, and throughput, and analyze their impact on processor performance.
2. Differentiate SRAM and DRAM by evaluating their latency, bandwidth, and throughput characteristics in cache and main memory design.
3. Compare virtual memory with paging and cache memory mechanisms in terms of access latency and system throughput.
4. Analyze the differences between NAND Flash and DRAM with respect to latency, bandwidth, and suitability for hierarchical memory systems.
5. Compare MRAM and RRAM technologies by examining their performance in terms of access time, throughput, and energy efficiency.

Code of Conduct:

*I **N. Tharun Sai (192472382)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

N. Tharun Sai

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided

Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

1. Compare L1, L2, and L3 Cache Levels in Terms of Latency, Bandwidth, and Throughput

Cache memory is a small and high-speed memory located close to the CPU. It stores frequently accessed instructions and data so that the processor does not have to access slower main memory repeatedly. Modern processors generally use multiple cache levels: L1, L2, and L3.

Parameter	L1 Cache	L2 Cache	L3 Cache
Size	Very small	Medium	Large
Location	Closest to CPU core	Close to CPU core	Usually shared
Latency	Very low	Low	Higher
Bandwidth	Very high	High	High
Throughput	Highest	High	Moderate–High
Cost per bit	High	High	Lower
Access frequency	Very frequent	Frequent	Less frequent
Typical purpose	Immediate CPU operations	Backup for L1	Backup for L1 and L2

L1 Cache: L1 is the fastest and smallest cache. It is normally located directly inside each CPU core. It is commonly separated into L1 Instruction Cache and L1 Data Cache. Because L1 is extremely close to the processor, it provides very low latency.

L2 Cache: L2 is larger than L1 but has slightly higher latency. Depending on processor architecture, it may be dedicated to a CPU core or shared by a small group of cores. When the required data is not found in L1, the processor checks L2.

L3 Cache: L3 is generally much larger than L1 and L2. It is often shared among multiple processor cores. It reduces the number of accesses that must reach DRAM.

Effect on Latency: The closer the cache is to the CPU, the lower its latency: $L1 < L2 < L3 < \text{DRAM}$.

Effect on Bandwidth: L1 generally provides the highest bandwidth because it is directly connected to the processor core. L2 also provides high bandwidth, while L3 provides high aggregate bandwidth but is often shared between CPU cores.

Effect on Throughput: Cache hits reduce the time the CPU spends waiting for data. Therefore, higher cache hit rates generally result in higher processor throughput.

Example: Consider a program repeatedly accessing the same array elements. The first access may bring the data into the cache. Subsequent accesses can be served from L1 or L2 instead of DRAM.

For example:

Without cache → CPU frequently waits for DRAM → lower throughput

With effective cache → CPU receives data quickly → higher throughput

Conclusion:

L1 provides the lowest latency and highest access speed, L2 provides a balance between speed and capacity, and L3 provides a larger shared cache that reduces accesses to DRAM. A well-designed L1–L2–L3 hierarchy improves CPU performance, reduces memory stalls, and increases overall system throughput.

2. Differentiate SRAM and DRAM by Evaluating Their Latency, Bandwidth, and Throughput

SRAM and DRAM are two important types of semiconductor memory. Although both are volatile, they are designed for different purposes.

Parameter	SRAM	DRAM
Full Form	Static RAM	Dynamic RAM
Storage	Flip-flop circuit	Capacitor and transistor
Speed	Very high	High
Latency	Very low	Higher than SRAM
Bandwidth	Very high	High
Throughput	Very high for cache access	High for main-memory access
Density	Lower	Higher
Cost	Expensive	Less expensive
Refresh	Not required	Required
Typical use	CPU cache	Main memory

SRAM: SRAM stores each bit using a circuit consisting of multiple transistors. It does not require periodic refresh while power is available. Its main advantages are very low access latency, high bandwidth, high-speed read/write operations, and suitability for frequently accessed data. Its main disadvantage is that it is expensive and physically large per bit.

Its main advantages are:

1. Very low access latency.
2. High bandwidth.
3. High-speed read and write operations.
4. No refresh operation.
5. Suitable for frequently accessed data.

DRAM: DRAM uses a transistor and capacitor to store each bit. The capacitor's charge gradually decreases, so the memory must periodically refresh its contents. Advantages include high storage density, lower cost per bit, and large capacity. The main disadvantage is greater latency and the need for refresh operations.

Advantages include:

1. High storage density.
2. Lower cost per bit.
3. Large capacity.
4. Suitable for main memory.

Latency Comparison: SRAM provides faster access because its stored state can be accessed without the refresh mechanism associated with DRAM. Therefore, SRAM has lower latency while DRAM has higher latency.

Bandwidth Comparison: SRAM can provide extremely high bandwidth for small amounts of data and is suitable for CPU cache operations. DRAM provides high bandwidth for large-capacity memory operations but generally has greater access latency.

Throughput: SRAM improves CPU throughput by supplying frequently needed data rapidly. DRAM provides the large working memory required for applications, but the processor may experience stalls when required data is not available in cache.

Why Both Are Required? Using only SRAM would make the computer extremely expensive and physically inefficient. Using only DRAM would increase processor waiting time. Therefore, computers combine them: CPU → SRAM Cache → DRAM → Storage.

Conclusion:

SRAM is preferred for high-speed cache memory, while DRAM is preferred for large-capacity main memory. Their combination provides an effective balance between latency, bandwidth, capacity, and cost.

3. Compare Virtual Memory with Paging and Cache Memory in Terms of Access Latency and System Throughput

Virtual memory and cache memory are different mechanisms, although both contribute to efficient memory utilization.

Parameter	Virtual Memory/Paging	Cache Memory
Main objective	Extend logical memory space	Reduce memory access latency
Managed by	Operating system + hardware	Hardware
Storage involved	RAM + secondary storage	SRAM
Access latency	High during page faults	Very low on cache hits
Unit	Page	Cache line/block
Main benefit	Larger address space	Faster CPU access
Effect on throughput	Can decrease during page faults	Usually increases
Main issue	Page fault	Cache miss

Virtual Memory: Virtual memory allows a system to provide each process with a large logical address space, even when physical RAM is limited. The operating system divides virtual memory into fixed-size pages, while physical RAM is divided into frames.

When the required page is not present in RAM, a page fault occurs. The operating system then has to bring the required page from secondary storage into RAM.

Effect of Page Faults: A page fault can introduce a very large delay compared with a normal memory access. If page faults happen frequently, CPU waiting increases, storage activity increases, and system throughput decreases.

If page faults happen frequently:

- CPU spends more time waiting.
- Disk/SSD activity increases.
- System throughput decreases.

- Application performance becomes slower.

Cache Memory: Cache memory is designed to reduce the time required for the CPU to access frequently used data. The processor generally searches memory in this order: L1 → L2 → L3 → DRAM.

A cache hit provides very fast access and allows the processor to continue execution quickly. A cache miss requires additional memory accesses and increases latency.

Throughput Comparison: Cache memory can significantly improve processor throughput because frequently used information is available close to the CPU. Virtual memory primarily provides memory isolation, larger logical address spaces, and efficient memory utilization.

Virtual memory primarily provides:

- Memory isolation.
- Larger logical address spaces.
- Efficient memory utilization.

Example:

With cache:

CPU → L1 cache → data found → fast execution.

Without cache:

CPU → DRAM → higher latency → CPU waits

With a page fault,

CPU → page table → page not in RAM → storage → page loaded → execution resumes.

Conclusion:

Cache memory is mainly a speed-enhancement mechanism, while virtual memory with paging is mainly a capacity and memory-management mechanism. Cache improves CPU throughput by reducing average memory-access latency, whereas excessive paging can reduce throughput because of expensive page faults.

4. Analyze the Differences Between NAND Flash and DRAM

NAND Flash and DRAM are semiconductor memory technologies with very different characteristics and applications.

Parameter	NAND Flash	DRAM
Volatility	Non-volatile	Volatile
Latency	High compared with DRAM	Low
Bandwidth	High in modern storage devices	Very high
Throughput	Good for storage workloads	High for active memory
Data retention	Retains data without power	Loses data without power
Refresh	Not required	Required
Density	Very high	High
Cost per bit	Lower	Higher
Main use	SSD/storage	Main memory

NAND Flash: NAND Flash is a non-volatile memory technology. It retains information even when power is removed. It is widely used in SSDs, USB flash drives, memory cards, smartphones, and embedded storage systems.

One important advantage of NAND Flash is high storage density. However, NAND Flash has significantly higher access latency than DRAM. It is therefore unsuitable as a direct replacement for fast CPU working memory in conventional systems.

It is widely used in:

- SSDs
- USB flash drives
- Memory cards
- Smartphones
- Embedded storage systems

Bandwidth and Throughput: Modern NAND-based SSDs can provide high sequential read/write bandwidth and good storage throughput. However, performance depends on NAND type, SSD controller, interface, parallelism, and workload.

DRAM: DRAM is used as the primary working memory of a computer. It is volatile, meaning data is lost when power is removed. DRAM provides much lower access latency than NAND Flash and is designed to support frequent processor memory accesses.

Hierarchical Position: A typical hierarchy is CPU → L1 → L2 → L3 → DRAM → NAND Flash. Moving downward generally provides increasing capacity, lower cost per bit, and increasing access latency.

Suitability: DRAM is suitable for operating-system working memory, application data, CPU working sets, and high-speed memory operations. NAND Flash is suitable for long-term storage, large files, operating-system storage, and persistent application data.

DRAM is suitable for:

- Operating system working memory
- Application data
- CPU working sets

NAND Flash is suitable for:

- Long-term storage
- Large files
- Operating system storage

Conclusion:

DRAM provides low latency and high memory performance, making it suitable for main memory. NAND Flash provides high density and non-volatility, making it suitable for persistent storage. Their different characteristics make both technologies important in a hierarchical memory system.

5. Compare MRAM and RRAM Based on Access Time, Throughput, and Energy Efficiency

MRAM and RRAM are advanced non-volatile memory technologies that are being investigated for future memory systems.

Parameter	MRAM	RRAM
Full Form	Magnetoresistive RAM	Resistive RAM
Data storage	Magnetic state	Resistance state
Volatility	Non-volatile	Non-volatile
Access time	Very fast	Fast
Latency	Low	Low
Throughput	High	High
Energy efficiency	High	Potentially very high
Endurance	High	Depends on implementation
Density	Good	High potential
Refresh	Not required	Not required

MRAM: MRAM stores information using magnetic properties rather than electrical charge. One important advantage is that it is non-volatile. Power OFF → Data remains stored. This makes MRAM useful in systems where fast startup and data persistence are important.

Access Time: MRAM can provide fast read/write operations and low access latency. Therefore, it can potentially be used in applications where both speed and persistence are required.

Throughput: MRAM can provide high throughput because it supports fast memory operations and does not require the refresh mechanism associated with DRAM.

Energy Efficiency: Because MRAM is non-volatile, it does not need continuous energy to maintain stored information. This can reduce standby power consumption.

PRAM: RRAM stores information using different resistance states of a material. A simplified representation is: low resistance → one logical state; high resistance → another logical state.

RRAM is attractive because of its potential for high density, low power, fast access, non-volatility, and compact memory cells.

RRAM is attractive because of its potential for:

- High density
- Low power
- Fast access
- Non-volatility
- Compact memory cells

Access Time: RRAM can offer fast switching and low access latency, although actual performance depends on the specific RRAM technology and implementation.

Energy Efficiency: RRAM is considered promising for low-energy memory applications. Its non-volatile nature eliminates the need for continuous refresh to retain data.

MRAM vs RRAM – Critical Analysis: MRAM has strong advantages in speed, endurance, and reliability, making it attractive for applications with frequent write operations. RRAM has strong potential in density and energy-efficient memory, making it attractive for future high-density memory architectures.

Conclusion:

MRAM and RRAM provide alternatives to traditional volatile memories. MRAM is particularly attractive when high speed, endurance, and non-volatility are required, while RRAM is promising for high-density and energy-efficient memory systems. Both can potentially complement conventional SRAM, DRAM, and NAND Flash in future hierarchical architectures.

Overall Summary

The five memory comparisons show that no single memory technology is best for every purpose. Hierarchical memory design balances latency, bandwidth, throughput, capacity, cost, and energy efficiency rather than relying on a single memory technology.

Memory	Main Strength	Typical Role
L1 Cache	Extremely low latency	Immediate CPU data
L2 Cache	Speed + larger capacity	Secondary cache
L3 Cache	Large shared cache	Reduce DRAM accesses
SRAM	Very high speed	Cache
DRAM	High capacity + speed	Main memory
NAND Flash	High density + persistence	Storage
MRAM	Speed + non-volatility	Advanced memory
RRAM	Density + energy efficiency potential	Future memory